past viewed in, 15–16
progress of astronomical events observed in, 15–16
range of meanings of, 10–11
cyclic universes, 3, 157, 169–70
gravity and, 66
issues in possibility of, 65–66
question of what survive from one cycle to another in, 66

dark energy, 155
cosmic expansion, 85–86, 88, 96, 97 (fig.), 109, 111 (fig.)
cosmic microwave background and, 187
cosmological constant and, 80–81, 90, 96, 97 (fig.), 104, 106–8, 109–10, 111 (fig.)
dominant energy condition and, 111
equation of state parameter for, 109, 114, 186
gravitational effect of pressure and, 108
Heat Death and, 114
measurement of, 105–6
negative pressure and, 108–9
past rate of expansion and, 115
phantom dark energy and, 110–12, 111 (fig.), 115
reasons for studying, 106
Type Ia supernovae explosions and, 125
dark matter, 67, 105, 155
Cosmic Dawn and, 48–49
evidence for, 67–68
gravitational lensing and, 68
Hubble Constant calculations and, 128
David, André, 196
de Beauvoir, Simone, 4
deceleration parameter, 72–74
De Luccia, Frank, 154
de Sitter Equilibrium, 100, 101
de Sitter space, 96, 97, 100
Dicke, Robert, 23, 27
Dirac, Paul, 199
distance ladder